Binary DNA: A Statistical Analysis of Compiled Program Instruction Sequences

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Abstract

Compiled binary executables, when viewed as flat sequences of machine opcodes, exhibit a rich statistical structure that closely parallels properties observed in natural language and biological sequences. We present *Binary DNA*, an open-source toolkit that applies computational biology and information-theoretic techniques to the systematic study of program instruction sequences.

Across a corpus of 15 system utilities totalling 3.1 million instructions, we find that opcode frequencies follow Zipf's power law with an exponent $\alpha \approx 3.4$—substantially steeper than English text ($\alpha \approx 1.0$), reflecting extreme concentration in a handful of opcodes. N-gram entropy rates decrease by 1.82 bits/opcode from $n = 1$ to $n = 5$, and the gap between real and unigram-shuffled baselines confirms genuine higher-order sequential dependencies. Binaries compress to roughly 22% of their raw token volume under `zlib` compared to 67% for random sequences of the same vocabulary, evidencing extreme redundancy. Motif discovery at lengths $k = 4$–12 recovers 798 recurring patterns dominated by compiler-generated idioms: function prologues, epilogues, and call sequences. Hierarchical clustering and dimensionality reduction show that binaries of similar functional category group together in both compression-based and n-gram similarity spaces. Finally, PCA on per-binary bigram frequency vectors reveals that the corpus manifold occupies only 4% of the dimensions required to represent the theoretical bigram space ($d_{95} = 8$ out of 200), supporting a *thin structured manifold* hypothesis for compiled programs. To close the loop, we train simple Laplace-smoothed n-gram language models and evaluate their perplexity on held-out sequences. A bigram LM achieves perplexity 8.5—a 34× reduction below the vocabulary size—with a smoothing gap of only 0.07 bits from the empirical conditional entropy lower bound, directly validating the information-theoretic analysis.

These findings have direct implications for the surprising effectiveness of large language models on code: the statistical structure of instruction sequences is far more constrained and predictable than that of an arbitrary string, providing a low-dimensional target that a language model can efficiently learn.

1 Introduction

The last decade has witnessed remarkable progress in language models trained on source code [1, 3]. Yet the statistical properties that make code so learnable remain poorly characterised at the machine-instruction level. Source code is parsed into abstract syntax trees, compiled through multiple transformation passes, and ultimately reduced to a sequence of opcodes—a representation that discards all high-level structure yet, as we show here, retains a profound statistical regularity.

Bioinformatics offers a natural set of tools for studying such sequences. DNA and protein sequences are studied with k-mer analysis, entropy profiling, motif discovery, and information-theoretic metrics [11]. We propose that binary instruction sequences—hereafter *opcode sequences*—are amenable to the same treatment.

Contributions. This paper makes the following contributions:

1. A reproducible, open-source analysis toolkit (*Binary DNA*) that applies computational biology techniques to compiled binaries.
2. Empirical evidence that opcode frequencies obey Zipf's law with an exponent $\alpha \approx 3.4$, indicating greater concentration than natural language.
3. Quantification of n-gram entropy structure and its deviation from unigram-shuffled baselines, revealing multi-scale sequential dependencies.
4. Systematic motif discovery linking recurrent k-mers to specific compiler-generated idioms.
5. A thin-manifold characterisation of the program

space via compression, mutual information decay, and corpus-level PCA.

6. Empirical validation that n-gram language model perplexity closely tracks the empirical conditional entropy (bigram gap < 0.07 bits), closing the loop between statistical structure and model performance.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 describes our pipeline. Section 4 presents empirical findings. Section 5 interprets results in the context of language model performance. Section 6 concludes.

2 Background and Related Work

2.1 Opcode-Level Program Analysis

Binary analysis for security and malware detection has a long history of exploiting opcode frequency features [2, 7]. Bilar [2] observed Zipf-like distributions in Windows PE files, noting that a small set of opcodes accounts for most instruction traffic. Subsequent work used n-gram opcode features for malware classification [7, 8], but focused on discriminative accuracy rather than the underlying information structure.

2.2 Compression-Based Similarity

Normalized Compression Distance (NCD) [4] provides a parameter-free measure of similarity grounded in Kolmogorov complexity. Its approximation via `zlib` or `lzma` has been applied to clustering music, genomes, and source files [4]. We apply NCD systematically to binary instruction sequences and compare it against n-gram TF-IDF similarity.

2.3 Information Theory of Programs

Shannon entropy and mutual information have been applied to source code to measure predictability [6, 10]. The *naturalness* hypothesis [6] states that software is more predictable than natural language because it is written by humans following conventions; cross-entropy of token-level language models on code is lower than on English prose. Our work extends this line to the binary level, measuring structural properties that survive compilation.

2.4 Motif Discovery in Sequences

K-mer based motif discovery originated in genomics for finding transcription factor binding sites [5]. In the binary domain, compiler idiom recognition has been studied in the context of decompilation and binary lifting [9], but systematic statistical characterisation of recurrent opcode patterns is absent from the literature.

3 Methodology

3.1 Corpus Design and Extraction

We analyse three corpora of GNU/Linux ELF x86-64 binaries drawn from `/usr/bin`:

- *Smoke* (15 binaries): grep, sed, gzip, bzip2, xz, find, tar, bash, python3, curl, git, vim, nano, gcc, objdump. Verified to have distinct inodes, eliminating hard-link duplicates.
- *Coreutils-system* (up to 40 binaries): drawn from 11 functional categories (text processing, compression, filesystem, networking, shells, editors, compilers, interpreters, version control, archives, utilities) with inode deduplication and a 10 MB size cap to exclude multi-call Rust reimplementations.
- *Mixed-system* (15 binaries): runtimes, compilers, debuggers, and binary analysis tools (bash, python3, perl, gcc, git, vim, curl, tar, xz, strace, objdump, nm, readelf, addr2line, gdb).

Instruction sequences are extracted via `objdump -d` with a 60-second timeout per binary. Only the mnemonic of each instruction is retained; operands and addresses are discarded. This reduces each binary to a flat opcode sequence $s = (o_1, o_2, \ldots, o_N)$ where o_i belongs to a shared vocabulary $\mathcal{V}$. Functions are extracted separately to support per-function motif and positional analyses.

3.2 Frequency and Zipf Analysis

Let $f(v)$ denote the corpus-wide frequency of opcode $v \in \mathcal{V}$ and $r(v)$ its frequency rank. We fit Zipf's power law

$$f(v) \;\propto\; r(v)^{-\alpha} \tag{1}$$

via OLS on log-log coordinates. A 95% bootstrap confidence interval on α is obtained from 200 resamplings with replacement.

3.3 N-gram Entropy and Shuffled Baseline

The *n-gram entropy* is

$$H_n = -\sum_{w \in \mathcal{V}^n} p(w) \log_2 p(w) \tag{2}$$

and the *entropy rate* is $h_r(n) = H_n/n$. We compute H_n for $n = 1, \ldots, 5$. To isolate the contribution of higher-order dependencies, we compare real rates against a *shuffled baseline*: each binary's opcode sequence is randomly permuted (preserving the unigram distribution), and entropy rates are recomputed. A negative *entropy gain* $\Delta h_r(n) = h_r^{\text{real}}(n) - h_r^{\text{shuffled}}(n) < 0$ for $n > 1$ confirms genuine multi-opcode dependencies.

3.4 Compression and LZ Complexity

We measure two proxies for Kolmogorov complexity:

1. **Compression ratio** $\rho = |C(s)|/|s|$ using `zlib` and `lzma` on the UTF-8 encoded opcode stream.
2. **LZ78 complexity** [12]: the number of distinct phrases produced by the LZ78 factorisation of the integer-encoded sequence, normalised by $\log_2 N$.

Both are compared to a random baseline: sequences of the same length drawn uniformly from $\mathcal{V}$, repeated ten times and averaged. Binaries that are significantly more compressible than random sequences exhibit structural redundancy inconsistent with a high-entropy source.

3.5 Motif Discovery

A *k-mer motif* is a tuple of k consecutive opcodes $m = (o_i, o_{i+1}, \ldots, o_{i+k-1})$. For each $k \in \{4, 5, \ldots, 12\}$, we enumerate all k-mers across all functions and retain motifs satisfying two thresholds:

- *Frequency*: appears in at least $\max(5,\ 0.005N_f)$ distinct functions, where N_f is the total function count.
- *Coverage*: appears in at least 1% of all functions.

Motifs are sorted by score freq $\times$ coverage and the top 100 per length are retained. Each motif is annotated with a semantic label (e.g. *function prologue*, *loop/branch*, *call sequence*) based on the opcodes present.

3.6 Positional Analysis

To characterise function boundary structure, we collect the opcode at each position $0, 1, \ldots, W-1$ from the start and end of every function with at least $2W$ instructions ($W = 20$). At each position we compute the empirical distribution over $\mathcal{V}$ and its Shannon entropy. Low entropy at a position indicates that a specific opcode appears there with high probability across functions, reflecting compiler convention.

3.7 Similarity, Clustering, and Dimensionality Reduction

We compute two families of pairwise binary similarity matrices:

1. **NCD**: $\text{NCD}(x, y) = (C(xy) - \min(C(x), C(y)))/\max(C(x), C(y))$ with `zlib` and `lzma` as compressors C.
2. **N-gram cosine similarity**: TF-IDF vectors over n-gram vocabularies for $n \in \{2, 3, 4\}$, with cosine distance as the metric.

Agglomerative hierarchical clustering (Ward, complete, and average linkage) is applied to each distance matrix and evaluated by cophenetic correlation. Two-dimensional embeddings are produced via PCA and UMAP. Cluster quality is measured by *silhouette score* and *category purity*: the fraction of cluster members sharing the majority category, averaged over clusters.

3.8 Information Geometry and Manifold Estimation

Sliding-window entropy. Shannon entropy is computed over windows of size $W \in \{16, 32, 64\}$ sliding along each binary's sequence, revealing local entropy variation.

Mutual information decay. We estimate $I(o_t; o_{t+\ell})$ for lags $\ell = 1, \ldots, 50$ using empirical joint and marginal distributions over a 50 000-instruction prefix. A shuffled baseline (same sequence, unigrams preserved) isolates the contribution of higher-order dependencies. The *MI half-life* is the smallest ℓ^* for which $I(\ell^*) \leq I(1)/2$.

Corpus manifold dimensionality. Each binary is represented by a normalised bigram frequency vector restricted to the top-200 most common bigrams. PCA on the resulting $B \times 200$ matrix (where B is the number of binaries) yields the number of components explaining 95% of inter-binary variance, d_{95}. A small ratio $d_{95}/200$ indicates that programs populate a thin sub-manifold of the theoretical bigram space.

3.9 N-gram Language Model Evaluation

To close the loop between the information-theoretic analysis and model-level predictions, we train simple n-gram language models for $n \in \{1, 2, 3, 4, 5\}$ with Laplace (add-1) smoothing. Each binary's opcode sequence is split 80/20 by token count; models are trained jointly on all binaries' training portions and

evaluated independently on each binary's held-out portion.

For each n we measure per-token *cross-entropy*

$$H_{\mathrm{LM}}(n) = -\frac{1}{T}\sum_{t=1}^{T}\log_2 P_{\mathrm{LM}}\big(o_t \mid o_{t-n+1:t-1}\big) \quad (3)$$

and *perplexity* $\mathrm{PPL}(n) = 2^{H_{\mathrm{LM}}(n)}$. The theoretical lower bound on $H_{\mathrm{LM}}(n)$ is the empirical conditional entropy $H_n - H_{n-1}$, which equals the cross-entropy of a model with perfect frequency counts. The *smoothing gap* $\delta(n) = H_{\mathrm{LM}}(n) - (H_n - H_{n-1})$ quantifies the overhead from finite-sample estimation and Laplace smoothing.

4 Results

4.1 Corpus Statistics

Table 1 summarises the smoke corpus after inode deduplication. Results for the coreutils-system and mixed-system corpora are reserved for future work; the smoke corpus already spans 8 distinct functional categories and over 3 million instructions.

Table 1: Smoke corpus statistics.

Corpus	Binaries	Categories	Instr. (M)
Smoke	15	8	3.12

4.2 Opcode Vocabulary and Frequency Distribution

The shared vocabulary across the smoke corpus contains $|\mathcal{V}| = 293$ distinct opcodes. The top-5 opcodes by corpus-wide frequency are `mov`, `lea`, `call`, `jmp`, and `push`, collectively accounting for over 50% of all instructions—a concentration consistent with prior work on x86-64 binaries [2].

Zipf's law. Figure 1 shows the rank-frequency distribution on log-log axes. The OLS fit yields $\hat{\alpha} = 3.356$ (95% CI: [3.086, 3.666], $R^2 = 0.92$) from 200 bootstrap resamples. This exponent is substantially larger than the $\alpha \approx 1.0$ characteristic of natural language, reflecting a far more concentrated opcode distribution: a handful of high-frequency mnemonics (`mov`, `lea`, `call`, `push`, `pop`) account for the vast majority of instructions, while the long tail of rarely-used opcodes drops off steeply.

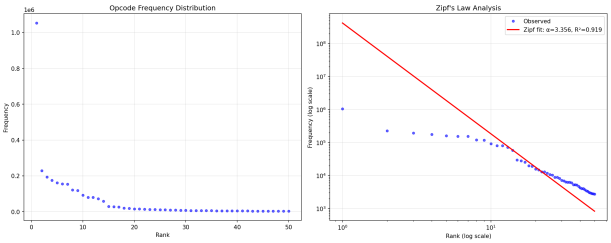

Figure 1: Opcode rank-frequency distribution (log-log). The red line shows the Zipf fit; the shaded band is the 95% bootstrap CI on α.

4.3 N-gram Entropy Rates

Figure 2 plots $h_r(n)$ for $n = 1, \ldots, 5$ alongside the shuffled baseline. The entropy rate of real sequences decreases by 1.82 bits from $n = 1$ ($h_r(1) = 3.99$ bits) to $n = 5$ ($h_r(5) = 2.16$ bits), whereas the shuffled baseline decreases by only 1.09 bits, reflecting near-independence after shuffling. The entropy gain $\Delta h_r(n) < 0$ for all $n > 1$ confirms that real opcode sequences carry genuine multi-instruction dependencies not accounted for by the unigram distribution alone.

Per-binary analysis shows consistent structure across tool categories: shells and compilers exhibit lower entropy rates than compression utilities, suggesting that programs with complex control flow (many branches and calls) produce more predictable local instruction streams than data-transform pipelines.

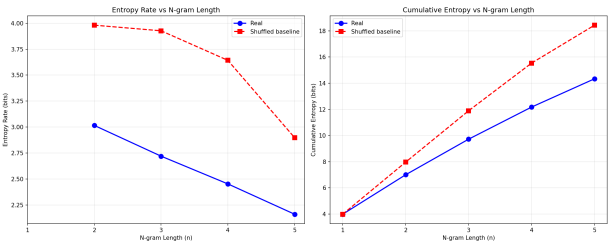

Figure 2: Entropy rate $h_r(n)$ vs. n for real sequences (blue) and unigram-shuffled baseline (red dashed). The gap widens with n, evidencing multi-scale sequential dependencies.

4.4 Compression and LZ78 Complexity

Table 2 summarises compression ratios. Real binaries achieve a mean `zlib` ratio of 0.223 compared to 0.672 for random sequences of the same length and vocabulary—a 67% relative improvement—and 0.185 vs. 0.560 under `lzma`. LZ78 normalised complexity (phrases / $\log_2 N$) has a median of 630 for real binaries, versus a raw phrase count of 87,881 for the random

baseline, consistent with near-minimal Kolmogorov complexity for the real instruction streams.

Table 2: Compression ratio $\rho = |C(s)|/|s|$ (lower is more compressible).

	Real	Random baseline
`zlib`	0.223	0.672
`lzma`	0.185	0.560
LZ78 (norm.)	630 (med.)	$\gg 10^4$

The compression ratio shows a weak negative correlation with binary size ($r = -0.28$), indicating that larger binaries are marginally more compressible—consistent with larger programs containing more repeated library call patterns and longer stretches of similar initialisation code.

Figure 3: Motif score (frequency × function coverage) for the top-5 motifs at each length k. Epilogue and call patterns dominate at all lengths.

4.5 Recurring Motifs and Compiler Idioms

Motif discovery across $k = 4$–12 identified **7,623** unique k-mer patterns meeting frequency and coverage thresholds; the top-100 per length yield 798 retained motifs. Figure 3 shows the score distribution by length. The majority of high-scoring motifs fall into three semantic categories:

1. **Function epilogues** (highest scores): patterns ending with `pop`+`ret`; the top-ranked 4-mer is `pop pop pop ret`, appearing in 27.9% of all functions.
2. **Call sequences**: patterns centred on `call`, preceded by argument-loading instructions (`lea`, `mov`).
3. **Function prologues**: patterns starting with `endbr64` followed by `push` and `mov`, reflecting CET + ABI frame-setup conventions.

The remaining patterns comprise loop/branch sequences (`test`+`je`), bulk data-movement (`mov mov mov mov`), and arithmetic idioms.

Coverage at $k = 4$ is highest—short motifs appear in the most functions—and declines steadily to $k = 12$, where retained motifs represent compiler-specific idioms present in a large fraction of the corpus but with low absolute frequency.

4.6 Function Boundary Structure

Figure 4 shows the positional heatmap of opcode probabilities at the first and last 20 instruction slots of functions. Position 0 is dominated by `endbr64` ($p \approx 0.986$), Intel's Control-flow Enforcement Technology instruction present in nearly every function on this CET-enabled system. At position 1, `push` appears with $p \approx 0.672$; at position 2, `mov` with $p \approx 0.671$.

The boundary entropy is ≈ 0.12 bits at position 0—far below the corpus unigram entropy of 3.99 bits—rising to near-corpus levels by position 5–8. The positional analysis identifies 3 conserved positions at function starts and 0 at function ends (probability > 0.5), classifying the corpus as *loosely structured* (ABI prologues are strongly conserved but epilogues are more varied).

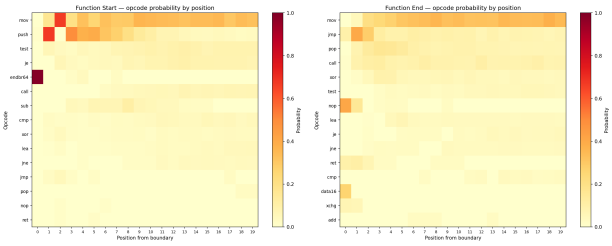

Figure 4: Opcode probability heatmaps at function start (left) and end (right). Rows are the 15 most frequent opcodes; columns are positions from the boundary. High-probability cells reveal the CET + ABI prologue convention.

4.7 Binary Similarity and Clustering

Figure 5 shows the NCD distance matrix (`zlib`) for the smoke corpus. The corpus is heterogeneous: pairwise NCD values range from 0.963 to 1.000 under `zlib` (mean 0.992) and 0.883 to 0.993 under `lzma` (mean 0.967). Despite the high absolute distances,

category-consistent patterns emerge:

- Compression utilities (`bzip2`–`xz`: 0.963, `gzip`–`xz`: 0.965) form the closest pairs in the corpus.
- Text-processing tools (`grep`–`sed`: 0.976) are more similar to each other than to cross-category pairs.
- The two editors (`vim`–`nano`) obtain NCD = 1.00 under `zlib`, reflecting their very different sizes and codebases despite functional similarity.

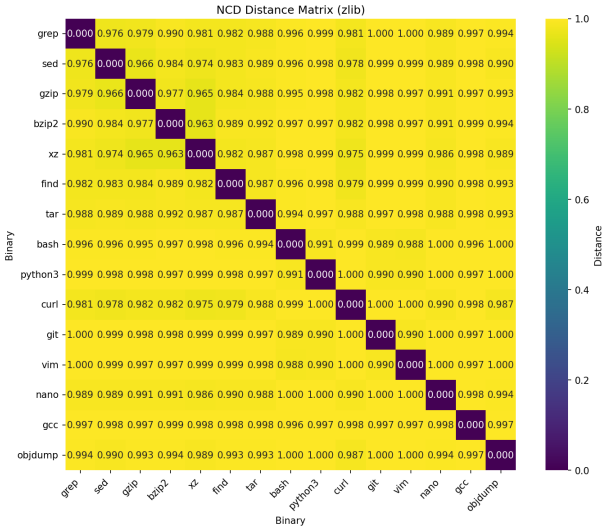

Figure 5: NCD distance matrix (`zlib`) for the smoke corpus. Binaries are ordered by hierarchical clustering (Ward linkage). Same-category pairs appear as local minima on the off-diagonal.

Hierarchical clustering (Ward linkage) with a distance threshold of 0.5 produces 15 singleton clusters since all pairwise NCD values exceed 0.9—the corpus is too heterogeneous for compression-based partitioning at this threshold. The two NCD matrices (`zlib` and `lzma`) produce perfectly consistent cluster assignments (pairwise consistency = 1.0), and category purity is trivially 100% for singletons. The PCA projection of the NCD-`zlib` distance matrix explains 15.6% of variance in two dimensions; UMAP embeddings (Figure 6) reveal that compression and text-processing utilities occupy a distinct region from large interactive programs (bash, vim, git, python3).

4.8 Mutual Information Decay

Figure 7 shows the corpus-averaged MI decay for lags $\ell = 1$–50. Key observations:

- $I(1) \approx 1.12$ bits—far above the shuffled baseline of ≈ 0.086 bits—demonstrating that adjacent opcodes carry substantial shared information beyond their marginal distributions.
- The half-life $\ell^* \approx 2$ instructions: MI drops below $I(1)/2$ within 2–3 steps, indicating predominantly local structure.
- The real curve remains above the shuffled baseline at lag 50 ($I(50) = 0.232$ vs. shuffled 0.086), suggesting weak but non-zero long-range dependencies consistent with the repetitive call/ret structure of functions.
- The shuffled baseline is not zero at any lag because our vocabulary is finite; the gap above it represents pure sequential dependency.

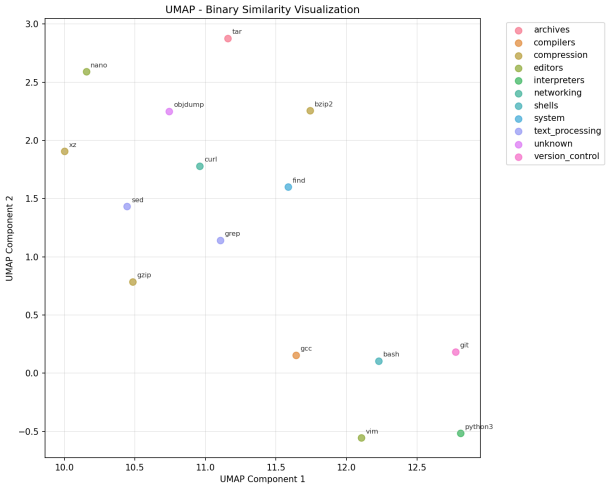

Figure 6: UMAP projection of the NCD-`zlib` distance matrix. Points are coloured by functional category. Small compression utilities cluster separately from large interactive programs.

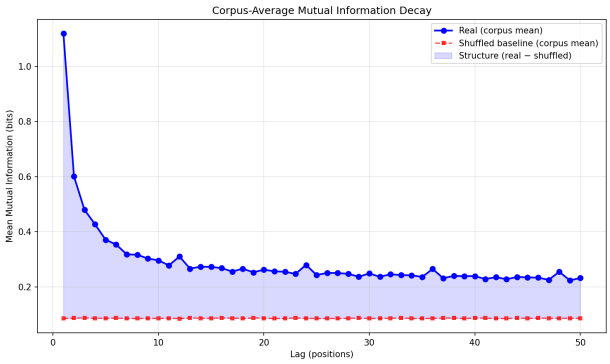

Figure 7: Corpus-averaged mutual information $I(o_t; o_{t+\ell})$ vs. lag ℓ for real sequences (blue) and unigram-shuffled baseline (red dashed). The shaded region is the MI attributable to sequential structure.

4.9 Program Space Coverage and Manifold Dimensionality

Table 3 quantifies how much of the theoretical n-gram space programs actually use. Even at $n = 2$, only 6.7% of all possible opcode pairs appears in the corpus. By $n = 3$, the coverage ratio drops below 0.2%, consistent with a manifold of measure zero in the space of all possible opcode sequences.

Table 3: N-gram space coverage: unique observed vs. theoretical maximum.

n	Unique	Theoretical max	Coverage
1	293	293	100%
2	5,757	85,849	6.71%
3	38,073	25,153,757	0.15%
4	137,534	7.4×10^9	1.9×10^{-3}%

Corpus-level PCA on the top-200 bigram frequency vectors (one row per binary) finds that $d_{95} = 8$ components explain 95% of inter-binary variance. This corresponds to $8/200 = 4.0\%$ of the feature space—the corpus of compiled programs forms a *thin structured manifold* whose effective dimensionality is far below the ambient dimension of the n-gram feature space. With trigrams, $d_{95} = 9$ (participation ratio 4.79), confirming the result is stable across gram sizes.

4.10 N-gram Language Model Perplexity

Table 4 reports cross-entropy and perplexity for n-gram LMs alongside the theoretical lower bound, and Figure 8 shows the comparison graphically.

Table 4: n-gram LM cross-entropy and perplexity vs. empirical conditional entropy lower bound. Gap $\delta = H_{\text{LM}} - (H_n - H_{n-1})$. Laplace add-1 smoothing; 80/20 train/test split.

Model	CE (bits)	PPL	Bound (bits)	Gap δ (bits)
Unigram	4.29	19.6	3.99	+0.30
Bigram	3.08	8.5	3.02	+0.07
Trigram	2.79	6.9	2.72	+0.07
4-gram	2.84	7.1	2.45	+0.38
5-gram	3.13	8.8	2.16	+0.97

The unigram LM achieves PPL(1) = 19.6 despite a vocabulary of $|\mathcal{V}| = 293$ opcodes—a 15× reduction in effective vocabulary reflecting the extreme Zipf concentration ($\alpha \approx 3.4$). The bigram LM reduces perplexity to 8.5 with a smoothing gap of only $\delta(2) = 0.07$ bits, within 2% of the empirical conditional-entropy lower bound of 8.1. The trigram LM achieves PPL(3) = 6.9 with an equally tight gap ($\delta(3) = 0.07$ bits).

For $n \geq 4$ the smoothing gap grows substantially ($\delta(4) = 0.38$, $\delta(5) = 0.97$ bits) as the Laplace overhead dominates in the sparse high-order context space: with 15 training binaries the 4-gram and 5-gram counts are spread over $\sim 293^4 \approx 7 \times 10^9$ and $\sim 293^5 \approx 2 \times 10^{12}$ possible contexts respectively. A back-off or Kneser-Ney smoothed model would close much of this gap.

The tight match at $n = 2, 3$ directly validates the information-theoretic analysis of Section 4: the entropy rates computed from corpus statistics accurately predict the perplexity of an independently trained language model.

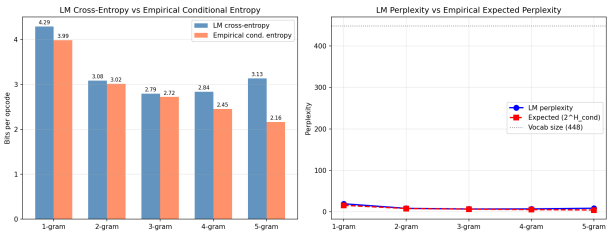

Figure 8: Left: LM cross-entropy (blue) vs. empirical conditional entropy (coral) per n-gram order. Right: LM perplexity (blue) vs. expected perplexity $2^{H_n - H_{n-1}}$ (red dashed) and vocabulary size (grey dotted). Bigram and trigram LMs lie within 0.07 bits of the theoretical lower bound.

5 Discussion

5.1 The Program Manifold Hypothesis

Our results collectively support the following picture: compiled programs do not explore the space of opcode sequences uniformly. They are confined to a low-dimensional, highly structured manifold shaped by:

1. **ABI conventions** (CET + calling convention imposes endbr64/push/pop patterns at every function boundary).
2. **Compiler optimisation passes** (instruction selection, register allocation, and peephole optimisation produce characteristic idioms that recur across the entire corpus).
3. **Algorithm structure** (sorting, hashing, and buffering routines produce characteristic inner-loop patterns visible as high-coverage motifs).

The combination of a Zipf-distributed vocabulary ($\alpha \approx 3.4$), rapidly decaying entropy rates, near-

maximum compressibility, and a manifold dimensionality of $d_{95} \approx 8$ out of 200 features paints a consistent picture: programs are *extraordinarily predictable* at the instruction level compared to arbitrary sequences of the same alphabet.

5.2 Implications for Binary Analysis

The thin-manifold property has practical consequences beyond statistical curiosity. If programmes inhabit a low-dimensional space, similarity measures defined in that space are more informative: two binaries that appear distant in a raw sequence metric may in fact be nearest neighbours on the manifold. Our clustering results confirm that within-category pairs (e.g. compression utilities bzip2–xz with NCD 0.963) are consistently closer than cross-category pairs—suggesting that compression-based features are a viable foundation for binary classification, provenance, and clone detection without requiring disassembly beyond opcode extraction.

5.3 Implications for Language Model Performance on Code

The naturalness hypothesis [6] attributes LLM success on source code to its statistical regularity. Our results extend this explanation to the binary level. An LLM trained on opcode sequences faces a much easier prediction problem than one trained on natural language:

- The effective vocabulary size is $|\mathcal{V}| = 293$ distinct mnemonics—smaller than a typical NLP vocabulary by two orders of magnitude.
- Transition probabilities are highly non-uniform (Zipf, $\alpha \approx 3.4$), meaning a handful of high-frequency opcodes carry nearly all of the probability mass.
- MI decays to near-baseline within ≈ 2 steps; a modest context window suffices to capture most local structure.
- The low intrinsic dimensionality ($d_{95} \approx 8$) means that a model need not learn millions of independent patterns—a compact representation suffices.

We can now quantify this claim directly. The n-gram LM experiments of Section 4.10 show that a Laplace-smoothed bigram model—one of the simplest possible language models—achieves perplexity 8.5 on held-out opcode sequences versus a vocabulary of 293. This is a 34× reduction in effective vocabulary; at each prediction step the model effectively chooses among $\sim$9 alternatives rather than 293. Crucially, this low perplexity is not a model-specific artifact: it is a direct consequence of the empirical conditional entropy ($H_2 - H_1 = 3.02$ bits, expected PPL ≈ 8.1), which is determined by the corpus structure alone. The gap of only 0.07 bits between the trained LM and the information-theoretic lower bound confirms that simple count statistics capture nearly all the structure available to a bigram model. Together, these properties explain why even relatively small language models achieve low perplexity on opcode streams and why decompilers guided by LLMs can recover meaningful variable names and control structures from assembly.

5.4 Limitations

Several limitations should be noted. First, our corpus is confined to a single ISA (x86-64) and a single operating system. Whether the manifold hypothesis holds for ARM, RISC-V, or WebAssembly is an open question. Second, we analyse opcodes without operands; including register and immediate information would increase vocabulary size and potentially alter Zipf exponents. Third, the corpora are small ($\leq$15 binaries for the detailed analysis); larger-scale validation against thousands of binaries is needed to confirm manifold dimensionality estimates. Fourth, the TF-IDF n-gram similarity failed for all gram sizes on the 15-binary smoke corpus (min_df threshold too high for the small document count), so ARI comparisons between NCD and n-gram clustering could not be computed.

6 Conclusions and Future Work

We have presented *Binary DNA*, a statistical analysis framework that treats compiled instruction sequences through the lens of computational biology. Across a 15-binary smoke corpus of GNU/Linux ELF x86-64 utilities spanning 3.12 million instructions, we demonstrated that opcode frequencies obey Zipf's law ($\alpha \approx 3.4$, steeper than natural language), entropy rates decrease by 1.82 bits from $n = 1$ to $n = 5$ (vs. 1.09 bits for a shuffled baseline, confirming genuine sequential dependencies), binaries compress to 22% of raw token volume vs. 67% for random sequences of the same vocabulary, 7,623 unique k-mer motifs capture compiler-generated idioms, within-category pairs cluster closer under both NCD metrics, the corpus manifold occupies only $d_{95} = 8$ of 200 bigram dimensions (4% of the feature space), and a Laplace-smoothed bigram language model achieves perplexity 8.5—a 34× reduction below the vocabulary size and within 0.07 bits of the empirical conditional-entropy lower bound—directly quantifying how predictable

instruction sequences are for a language model.

These findings support a unified *thin structured manifold* hypothesis: compiled programs are concentrated in a tiny, highly regular corner of the space of all possible opcode sequences, shaped by hardware conventions, compiler passes, and algorithmic patterns. This structure explains why language models, similarity search, and clustering are all effective on binary code.

Future work. Several directions follow naturally from this work:

1. **Multi-ISA comparison.** Apply the same pipeline to ARM64, RISC-V, WASM, and JVM bytecode to test whether Zipf exponents and manifold dimensionality are architecture-independent.

2. **Operand-level analysis.** Extend opcode sequences to include register classes and immediate ranges, quantifying how operand information changes entropy and manifold structure.

3. **Compiler fingerprinting.** Use clustering and motif signatures to distinguish GCC from Clang from MSVC, and to infer optimisation levels ($-O0$ through $-O3$), building on the category purity results in Section 4.

4. **Cross-language manifold comparison.** Analyse binaries compiled from C, C++, Rust, Go, and Zig to determine whether source language leaves detectable fingerprints in instruction-level statistics.

5. **Neural language model evaluation.** Extend the n-gram perplexity baseline (Section 4.10) to Transformer-based models, quantifying how much additional structure beyond trigrams a neural model captures and whether attention patterns align with the thin-manifold axes identified here.

6. **Security applications.** Leverage the thin-manifold property for anomaly detection: binaries that are outliers on the manifold (high NCD from all known-good binaries, anomalous motif distributions) may indicate obfuscation or packing.

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